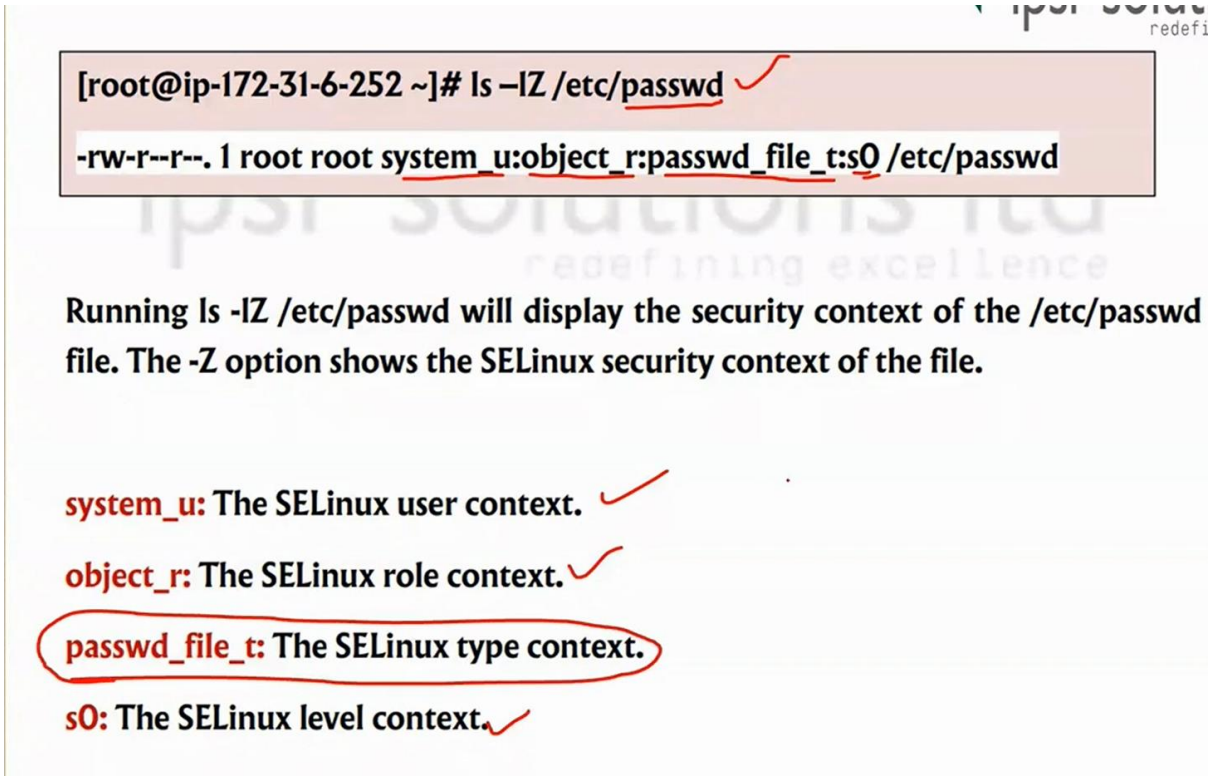
SELINUX

Selinux(security enhance linux)

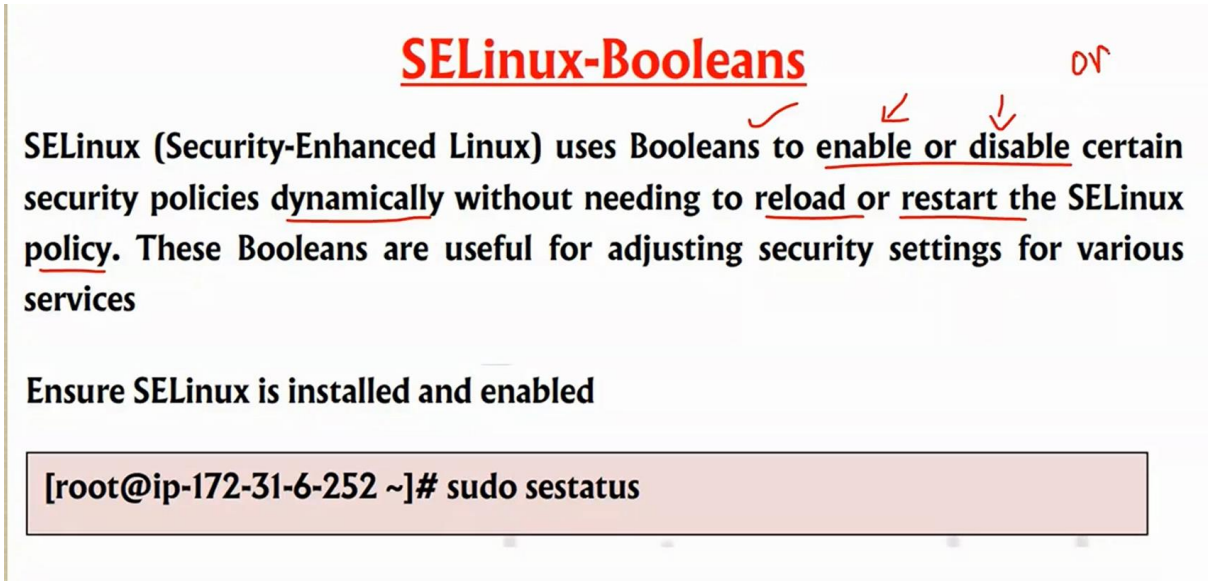
- Selinux is a default security in redhat
- For example os level security is firewall
- Service related matters selinux active
- We can check se linux mode using this command #sestatus
- Selinux conf file is /etc/selinux/config
- 3 type od mode
- Enforcing (it's a secure mode)
- Permissive(this mode mainly used for testing purpose)
- Disabled mode (completely we can disable se linux here)
- Setenforce o enter for permissive after edit conf file
- Setenforce 1
- Sestatus for check mode



- We can temperorly change a file context using this command
- #chcon -t httpd_sys_context_t file.txt
- Selinux related things like context,labels for its we using command is semanage

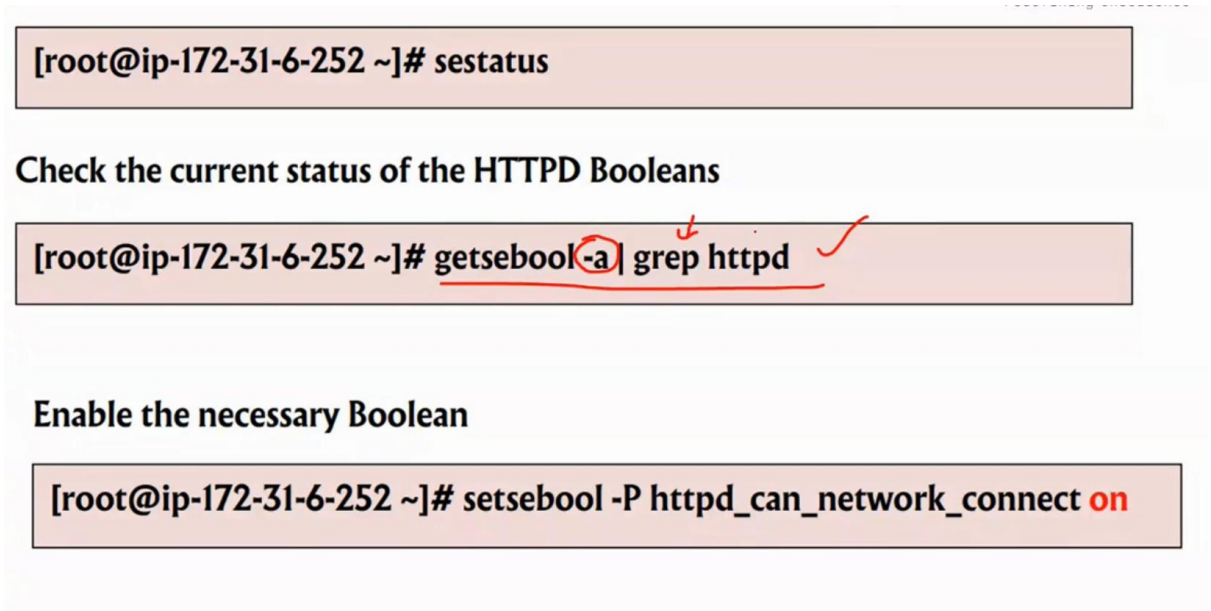
- #semanage fcontext -a -t httpd_sys_content_t “/web(/.*)?”
- This command for changing selinux context type
- After this command we need to restore that directory for change all files inside their
- #restorecon -FR -v /web
- ls -lZ (after this command we can see the all file context changed inside that dir)

SElinux-Booleans



For example we run a web application apache related we want to external database or need to connect another webserver here by default se linux prevent it.

Here we can use Boolean to prevent that for Boolean no restart or reboot thongs is needed just off and on that enough



getsebool is used in Linux to check the current status of SELinux Booleans.

#getsebool -a | grep httpd

For our case we want to enable httpd_can_network_connect → off ->on

For enable this we setsebool -p httpd_can_network_connect on

After that we can check status

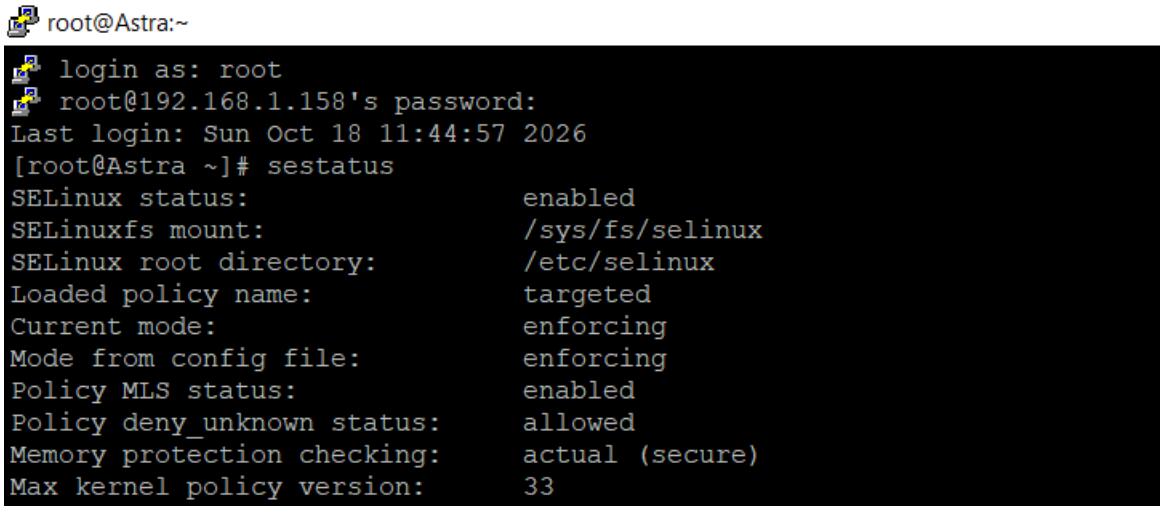
getsebool httpd_can_network_connect

another example we developed a web application that need to send email in this case also selinux prevent it

so on that using getsebool for use it .this like.

Practical_screenshot

Selinux status :-



- **selinux mode changing:-**

```
[root@Astra ~]# vim /etc/selinux/config
[root@Astra ~]# setenforce 0
[root@Astra ~]# sestatus
SELinux status:                 enabled
SELinuxfs mount:                /sys/fs/selinux
SELinux root directory:         /etc/selinux
Loaded policy name:             targeted
Current mode:                   permissive
Mode from config file:          permissive
Policy MLS status:              enabled
Policy deny_unknown status:     allowed
Memory protection checking:     actual (secure)
Max kernel policy version:      33
[root@Astra ~]# vim /etc/selinux/config
[root@Astra ~]# setenforce 1
[root@Astra ~]# sestatus
SELinux status:                 enabled
SELinuxfs mount:                /sys/fs/selinux
SELinux root directory:         /etc/selinux
Loaded policy name:             targeted
Current mode:                   enforcing
Mode from config file:          enforcing
Policy MLS status:              enabled
Policy deny_unknown status:     allowed
Memory protection checking:     actual (secure)
Max kernel policy version:      33
```

Creating a dir /janeeb and janeeb.html file

```
[root@Astra dev]# mkdir /janeeb
[root@Astra dev]# cd
[root@Astra ~]# cd /janeeb
[root@Astra janeeb]# ls -lZ
total 0
[root@Astra janeeb]# vim janeeb.html
```

Changing /janeeb dir file context to httpd_sys_content_t to enable selinux security

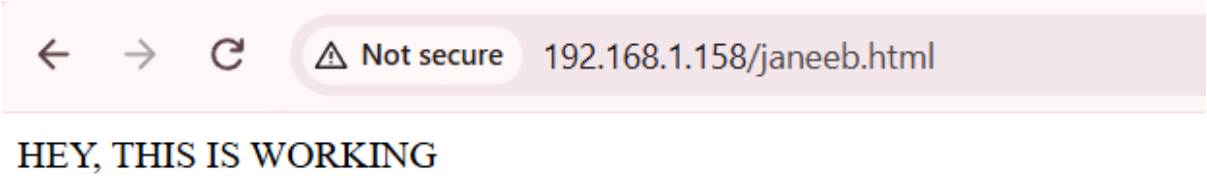
```
[root@Astra ~]# semanage fcontext -a -t  httpd_sys_content_t "/janeeb(/.*)?"
[root@Astra ~]# ls -lZ /janeeb
total 4
-rw-r--r--. 1 root root unconfined_u:object_r:default_t:s0 22 Oct 18 11:55 janeeb.html
[root@Astra ~]# restorecon -FR -v /janeeb
Relabeled /janeeb from unconfined_u:object_r:default_t:s0 to system_u:object_r:httpd_sys_cont
ent_t:s0
Relabeled /janeeb/janeeb.html from unconfined_u:object_r:default_t:s0 to system_u:object_r:ht
tpd_sys_content_t:s0
[root@Astra ~]# ls -lZ /janeeb
total 4
-rw-r--r--. 1 root root system_u:object_r:httpd_sys_content_t:s0 22 Oct 18 11:55 janeeb.html
```

After that restorecon -FR -v /janeeb

ip addr

```
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 10
00
    link/ether 00:0c:29:f9:cd:ac brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 192.168.1.158/24 brd 192.168.1.255 scope global dynamic noprefixroute ens160
       valid_lft 85054sec preferred_lft 85054sec
    inet6 2401:4900:8fdd:8258:20c:29ff:fef9:cdac/64 scope global dynamic noprefixroute
       valid_lft 43447sec preferred_lft 43447sec
    inet6 fe80::20c:29ff:fef9:cdac/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
[root@Astra ~]# ^C
```

http://192.168.1.158/janeeb.html



HEY, THIS IS WORKING

we use getsebool to check SELinux Boolean values.

```
       valid_lft forever preferred_lft forever
[root@Astra ~]# getsebool -a | grep httpd
httpd_anon_write --> off
httpd_builtin_scripting --> on
httpd_can_check_spam --> off
httpd_can_connect_ftp --> off
httpd_can_connect_ldap --> off
httpd_can_connect_mythtv --> off
httpd_can_connect_zabbix --> off
httpd_can_manage_courier_spool --> off
httpd_can_network_connect --> off
httpd_can_network_connect_cobbler --> off
httpd_can_network_connect_db --> off
httpd_can_network_memcache --> off
httpd_can_network_relay --> off
httpd_can_sendmail --> off
httpd_dbus_avahi --> off
httpd_dbus_sssd --> off
httpd_dontaudit_search_dirs --> off
httpd_enable_cgi --> on
httpd_enable_ftp_server --> off
httpd_enable_homedirs --> off
httpd_execmem --> off
```

This like we on Boolean value

```
[root@Astra ~]# setsebool -p httpd_can_network_connect on
setsebool: invalid option -- 'p'

Usage:  setsebool [ -NPV ] boolean value | bool1=val1 bool2=val2...

[root@Astra ~]# setsebool -p httpd_can_network_connect on^C
[root@Astra ~]# setsebool -P httpd_can_network_connect on
```

```
httpd_can_connect_ldap --> off
httpd_can_connect_mythtv --> off
httpd_can_connect_zabbix --> off
httpd_can_manage_courier_spool --> off
httpd_can_network_connect --> on
httpd_can_network_connect_cobbler --> off
httpd_can_network_connect_db --> off
httpd_can_network_memcache --> off
httpd_can_network_relay --> off
```